

Existence of Solutions Algebraic Equations from Existence of Solutions of Ordinary Differential Equations (常微分方程式の解の存在定理からの代数学の基本定理)

Masatoshi Ohrui

Existence theorem of solutions of ordinary differential equations (Cauchy-Lipschitz theorem) follows without existence of algebraic equations. We can prove Existence of solutions of algebraic equations from existence of solutions of ordinary differential equations. (常微分方程式の解の存在定理(コーシー-リプシッツの定理)の証明に代数学の基本定理は使わない. 代数学の基本定理を常微分方程式の解の存在定理から証明できる).

An initial value problem $P(D)u(x) = 0, x \in I, u^{(k)}(0) = u_{0k} (k = 0, \dots, n-1)$ has an unique solution $u \in C^1(\bar{J})$ from corollary of Cauchy-Lipschitz theorem ([1]): if f is Lipschitz continuous on a closed interval I , then an initial value problem $u^{(n)}(x) = f(x, u, u', \dots, u^{(n-1)}), u^{(k)}(0) = u_{0k} (k = 0, \dots, n-1)$ has an unique solution on a closed interval J .

Linear ordinary differential equations with constant coefficients on bounded and closed interval I , an initial value problem $P(D)u(x) = 0, x \in I, u^{(k)}(0) = u_{0k} (k = 0, \dots, n-1)$ has an solution. To prove it, let $0 = P(x) = \sum_{i=0}^n a_i x^i \Rightarrow x^n = Q(x) = \sum_{i=0}^{n-1} -(a_i/a_n)x^i$. Since $Q(x, x_1, \dots, x_{n-1}) = \sum_{i=0}^{n-1} -(a_i/a_n)x_i^i (x_0 = x)$ is Lipschitz continuous, so $u^{(n)}(x) = Q(D)u(x), u^{(k)}(0) = u_{0k}$ has a solution on the neighborhood $0 \in J = \bar{J} \subset I$. To prove it, put $v = u^{(n-1)}, f(x, u, \dots, u^{(n-1)}) = Q(D)u(x)$, use mathematical induction and Cauchy-Lipschitz theorem to $v'(x) = f(x, u, \dots, v), v(0) = u^{(n-1)}(0)$.

u has an extension $\bar{u} \in \mathcal{S}$. By Fourier transform, we obtain $P(i\xi)\mathcal{F}\bar{u}(\xi) = 0$.
 $P(i\xi)\mathcal{F}\bar{u}(\xi) = 0 \iff P(D)\bar{u}(x) = 0$ ([2]). We can choose $\bar{u}$ as $\exists \xi, \mathcal{F}\bar{u}(\xi) \neq 0$, so
 we have an existence of ξ such that $P(i\xi) = 0$.

References: [1]石原茂-矢野健太郎, 解析学概論, 裳華房, 2020; [2]垣田高雄-柴田良弘, ベクトル解析から
 流体へ, 日本評論社, 2024.

